Exercise C - II

1. Let f be a one–one function with domain {x,y,z} and range {1,2,3}. It is given that exactly one of the following statements is true and the remaining two are false. $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. Determine $f^{-1}(1)$

   f एकैकी फलन है। जिसका प्रान्त {x,y,z} व परिसर {1,2,3} है दिये गये कथनों में से मात्र एक सत्य है $f(x) = 1$; $f(y) \neq 1$; $f(z) \neq 2$. $f^{-1}(1)$ ज्ञात कीजिए।

Ans. $f^{-1}(1) = y$

Sol. Case-I

S-I is true, S-II & S-III are false.

$f(x) = 1$

$f(y) = 1$

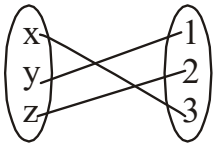
$f(z) = 2$

Not one-one.

Case-II

If S-II is true & S-I & S-III are false.

$\underbrace{f(x) \neq 1, f(y) \neq 1}_{\text{Not one-one}}$, $f(z) = 2$



Case-III

S-III is true, S-II, S-I false.

$f(x) \neq 1, f(y) = 1, f(z) = 2$

$f^{-1}(1) = y$

2. Let $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ be a polynomial such that $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ and $P(6) = 6$ then find the value of $P(7)$.

   $P(x) = x^6 + ax^5 + bx^4 + cx^3 + dx^2 + ex + f$ बहुपद है जहाँ $P(1) = 1$; $P(2) = 2$; $P(3) = 3$; $P(4) = 4$; $P(5) = 5$ व $P(6) = 6$ तो $P(7)$ का मान ज्ञात कीजिए।

Ans. 727

Sol. $P(x) = x \ \forall \ x = \{1,2,3,4,5,6\}$

$P(x) - x = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5)(x - 6)$

$P(7) - 7 = 720$

$P(7) = 727$

3. Let a and b be real numbers and let $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$. If $f\left(\log_{10}(\log_3 10)\right) = 5$ then find the value of $f\left(\log_{10}(\log_{10} 3)\right)$.

a व $b \in R$ तथा $f(x) = a\sin x + b\sqrt[3]{x} + 4, \forall x \in R$ यदि $f\left(\log_{10}(\log_3 10)\right) = 5$ तो $f\left(\log_{10}(\log_{10} 3)\right)$ का मान ज्ञात कीजिए।

Ans. 3

Sol. $f(\log_{10}\log_3 10) = 5$

$$\Rightarrow f(-x) = -a\sin x - bx^{\frac{1}{3}} + 4$$

$$f(x) = a\sin x + bx^{\frac{1}{3}} + 4$$

$$\Rightarrow f(x) + f(-x) = 8$$

$$\Rightarrow f(\log_{10}\log_3 10) + f(-\log_{10}\log_3 10) = 8$$

$$\Rightarrow f(\log_{10}\log_{10} 3) = 3$$

4. Let f be a real valued invertible function such that $f\left(\frac{2x-3}{x-2}\right) = 5x - 2, x \neq 2$. Find $f^{-1}(13)$.

माना f कोई वास्तविक मान का व्युत्क्रमणीय फलन है जिसके लिये $f\left(\frac{2x-3}{x-2}\right) = 5x - 2, x \neq 2$ है। $f^{-1}(13)$ मान निकालिये।

Ans. 3

Sol. $f\left(\frac{2x-3}{x-2}\right) = 5x - 2$

$$f^{-1}(5x-2) = \frac{2x-3}{x-2}$$

Put $x = 3$

$$f^{-1}(13) = \frac{6-3}{3-2} = 3$$

5. Let [x] = the greatest integer less than or equal to x. If all the values of x such that the product $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right]$ is prime, belongs to the set $[x_1, x_2) \cup [x_3, x_4)$, find the value of $x_1^2 + x_2^2 + x_3^2 + x_4^2$.

गुणनफल $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right]$ अभाज्य है। x का मान $[x_1, x_2) \cup [x_3, x_4)$ में है तो $x_1^2 + x_2^2 + x_3^2 + x_4^2$ का मान ज्ञात करें। [.] महत्तम पूर्णांक फलन है।

Ans. 11

Sol. $\left[x-\frac{1}{2}\right]\left[x+\frac{1}{2}\right] =$ prime

Case-I $1 \times 2 = 2$

$\left[x+\frac{1}{2}\right]=2$ $\qquad$ $\left[x-\frac{1}{2}\right]=1$

$x+\frac{1}{2}\in[2,3)$ $\qquad$ $x-\frac{1}{2}\in[1,2)$

$x\in\left[\frac{3}{2},\frac{5}{2}\right)$ $\qquad$ $x\in\left[\frac{3}{2},\frac{5}{2}\right)$

Case-II $(-2) \times (-1) =$ prime

$\left[x+\frac{1}{2}\right]=-1$ $\qquad$ $\left[x-\frac{1}{2}\right]=-2$

$x+\frac{1}{2}\in[-1,0)$ $\qquad$ $x-\frac{1}{2}\in[-2,-1)$

$x\in\left[-\frac{3}{2},-\frac{1}{2}\right)$ $\qquad$ $x\in\left[-\frac{3}{2},-\frac{1}{2}\right)$

$$x_1^2 + x_2^2 + x_3^2 + x_4^2 = \left(\frac{3}{2}\right)^2 + \left(\frac{5}{2}\right)^2 + \left(\frac{-3}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 = \frac{44}{4} = 11$$

6. Suppose p(x) is a polynomial with integer coefficients. The remainder when p(x) is divided by x – 1 is 1 and the remainder when p(x) is divided by x – 4 is 10. If r (x) is the remainder when p(x) is divided by (x – 1) (x – 4), find the value of r (2006).

बहुपद p(x) के गुणांक पूर्णांक है। x – 1 से विभाजित करने पर शेषफल 1 व x – 4 से विभाजित करने पर शेषफल 10 है। (x – 1)(x – 4) से P(x) को विभाजित करने पर r(x) शेषफल है तो r (2006) ज्ञात कीजिए।

Ans. 6016

Sol. $p(x) = (x-1)\,Q_1(x) + 1 \Rightarrow P(1) = 1$

$p(x) = (x-4)\,Q_2(x) + 10 \Rightarrow P(4) = 10$

$P(x) = (x-1)(x-4)\,Q(x) + ax + b$

$P(1) = a + b = 1$

$P(4) = 4a + b = 10$

$a = 3,\ b = -2$

$r(x) = 3x - 2$

$r(2006) = 6016$

7. Prove that the function defined as, $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$ f(x) is odd as well as even. (where {x} denotes the fractional part function)

सिद्ध कीजिये कि $f(x) = \begin{cases} e^{-\sqrt{|\ln\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\ln\{x\}|}}} & \text{where ever it exists} \\ \{x\} & \text{otherwise, then} \end{cases}$ जहाँ परिभाषित है, सम व विषम दोनों है।

Ans.

Sol. $f(x) = \begin{cases} e^{-\sqrt{|\ell n\{x\}|}} - \{x\}\sqrt{\frac{1}{|\ell n\{x\}|}} & \text{wherever it exists} \\ \{x\} & \text{otherwise} \end{cases}$

Let

$y = e^{-\sqrt{|\ell n\{x\}|}}$ $\qquad$ $z = \{x\}^{\sqrt{\frac{1}{|\ell n\{x\}|}}}$

$\ell ny = -\sqrt{|\ell n\{x\}|}$ $\qquad$ $\ell nz = \sqrt{\frac{1}{|\ell n\{x\}|}}\ell n\{x\}$

$$= -\sqrt{|\ell n\{x\}|}$$

$\Rightarrow y = z \forall x \in R - z$

$$f(x) = \begin{cases} 0 & x \in R - z \\ 0 & x \in z \end{cases}$$

f(x) is both even and odd.

8. In a function $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\left|\sqrt{2}\ \sin\left(\pi\left(x + \frac{1}{4}\right)\right)\right|\right) = 4\cos^2\frac{\pi x}{2} + x\cos\frac{\pi}{x}$

Prove that (i) $f(2) + f(1/2) = 1$ and (ii) $f(2) + f(1) = 0$

फलन $2f(x) + xf\left(\frac{1}{x}\right) - 2f\left(\left|\sqrt{2}\ \sin\left(\pi\left(x + \frac{1}{4}\right)\right)\right|\right) = 4\cos^2\frac{\pi x}{2} + x\cos\frac{\pi}{x}$

के लिये सिद्ध कीजिए कि (i) $f(2) + f(1/2) = 1$ तथा (ii) $f(2) + f(1) = 0$

Ans.

Sol. put $x = 1$

$2f(1) + 1f(1) - 2f(|\sqrt{2}\ \sin\frac{5\pi}{4}|) = -1$

$\Rightarrow \quad 3f(1) - 2f(1) = -1 \quad \Rightarrow \quad f(1) = -1$

Now put $x = 2$

$$2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4\cos^2\pi + 2\cos\frac{\pi}{2}$$

$$\Rightarrow \quad 2f(2) + 2f\left(\frac{1}{2}\right) - 2f(1) = 4$$

$$\Rightarrow \quad f(2) + f\left(\frac{1}{2}\right) = 1 \qquad .....(i)$$

Now put $x = 1/2$ we get

$$4f\left(\frac{1}{2}\right) + f(2) = 1 \quad .....(ii)$$

from (i) and (ii)

$$f\left(\frac{1}{2}\right) = 0 \text{ and } f(2) = 1$$

9. A function f, defined for all $x, y \in R$ is such that $f(1) = 2$; $f(2) = 8$ and $f(x+y) - kxy = f(x) + 2y^2$, where k is some constant. Find f(x) and show that $f(x+y)\, f\left(\frac{1}{x+y}\right) = k$ for $x + y \neq 0$.

फलन f, $x, y \in R$ इस प्रकार है कि $f(1) = 2$; $f(2) = 8$ तथा $f(x+y) - kxy = f(x) + 2y^2$, जहाँ k नियतांक है। f(x) ज्ञात करें और सिद्ध कीजिये कि : $f(x+y)\, f\left(\frac{1}{x+y}\right) = k$ for $x + y \neq 0$.

Ans. $f(x) = 2x^2$

Sol. $f(x + y) - kxy = f(x) + 2y^2$

$x = y = 1$

$8 - k = 2 + 2$

$k = 4$

$x = 0, y = 1$

$f(1) = f(0) + 2$

$f(0) = 0$

$x = 0, y = y$

$f(y) = 2y^2$

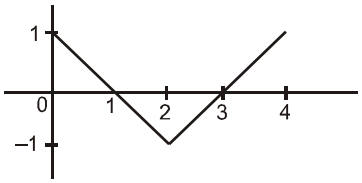
10. If $f(x) = -1 + |x-2|$, $0 \le x \le 4$
$g(x) = 2 - |x|$, $-1 \le x \le 3$
Then find fog(x) and gof(x). Draw rough sketch of the graphs of fog(x) and gof(x).

यदि $f(x) = -1 + |x-2|$, $0 \le x \le 4$
$g(x) = 2 - |x|$, $-1 \le x \le 3$
fog(x) व gof(x) ज्ञात कीजिए। ग्राफ भी खीचियें।

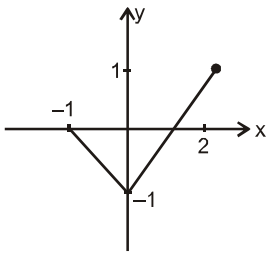
Ans. $fog(x) = \begin{matrix} -(1+x) & , & -1 \le x \le 0 \\ x-1 & , & 0 < x \le 2 \end{matrix}$; $gof(x) = \begin{matrix} x+1 & , & 0 \le x < 1 \\ 3-x & , & 1 \le x \le 2 \\ x-1 & , & 2 < x \le 3 \\ 5-x & , & 3 < x \le 4 \end{matrix}$;

$fof(x) = \begin{matrix} x & , & 0 \le x \le 1 \\ 4-x & , & 3 \le x \le 4 \end{matrix}$; $gog(x) = \begin{matrix} -x & , & -1 \le x \le 0 \\ x & , & 0 < x \le 2 \\ 4-x & , & 2 < x \le 3 \end{matrix}$

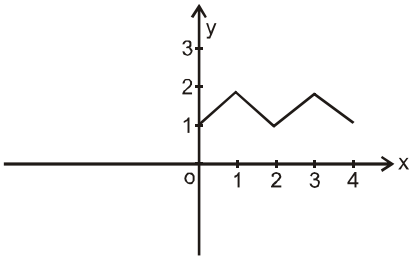
Sol. $f(x) = -1 + |x-2|$



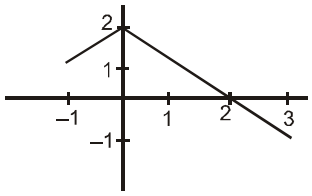
graph of fog(x) is



graph of gof(x) is



$g(x) = 2 - |x|$



$fog(x) = -1 + |x|$ $\quad -1 \le x \le 2$

$gof(x) = 2 - |-1 + |x-2|| \quad 0 \le x \le 4$

$fof(x) = -1 + |-1 + |x-2| - 2|$

$= -1 + | |x-2| - 3| \quad 0 \le x \le 1$, and $3 \le x \le 4$

$gog(x) = 2 - |2 - |x|| \quad x \in [-1, 3]$

11. If for all real values of u and v, $2f(u)\cos v = f(u+v) + f(u-v)$, prove that, for all real values of x.

(i) $f(x) + f(-x) = 2a\cos x$

(ii) $f(\pi - x) + f(-x) = 0$

(iii) $f(\pi - x) + f(x) = -2b\sin x$. Deduce that $f(x) = a\cos x - b\sin x$, a, b are arbitrary constants.

यदि u तथा v के लिए सभी वास्तविक मान, $2f(u)\cos v = f(u+v) + f(u-v)$, सिद्ध कीजिये कि x के वास्तविक मान होंगे।

(i) $f(x) + f(-x) = 2a\cos x$

(ii) $f(\pi - x) + f(-x) = 0$

(iii) $f(\pi - x) + f(x) = -2b\sin x$. दिखाइये कि $f(x) = a\cos x - b\sin x$, a, b नियतांक है।

Ans.

Sol. $2f(u)\cos v = f(u+v) + f(u-v)$

(i) put $u = 0$, $v = x$

$\Rightarrow f(x) + f(-x) = 2f(0)\cos x$

$f(0) =$ any constant a

$f(x) + f(-x) = 2a\cos x$ _____(1)

(ii) $v = \frac{\pi}{2}$, $u = \frac{\pi}{2} - x$

$$2f\left(\frac{\pi}{2}-x\right)(0)=f(\pi-x)+f(-x)$$

$$\Rightarrow f(\pi-x)+f(-x)=0 \qquad \text{_____(2)}$$

(iii) $v=\frac{\pi}{2}-x$, $u=\frac{\pi}{2}$

$$2f\left(\frac{\pi}{2}\right)\sin x=f(\pi-x)+f(x)$$

$$\Rightarrow f(\pi-x)+f(x)=-2b\sin x \text{ where } b=-f\left(\frac{\pi}{2}\right) \qquad \text{_____(3)}$$

(1) + (3) – (2)

$2f(x) = 2a\cos x - 2b\sin x$

$$\Rightarrow f(x)=a\cos x-b\sin x$$

12. A function f: $R \to R$ where R is the set of real numbers is defined by $f(x)=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$. Find the interval of values of α for which f is onto. Is the function one–to–one for $\alpha = 3$? Justify your answer.

फलन $f: R \to R$ जहाँ R सभी वास्तविक संख्याओं का समुच्चय है, $f(x)=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$. α के मानों का अन्तराल ज्ञात कीजिए, यदि f आच्छादक हो। यदि $\alpha = 3$, तो क्या फलन f एकैकी होगा।

Ans. [2, 14]; for $\alpha = 3$, f is not one–one

Sol. $y=\frac{\alpha x^2+6x-8}{\alpha+6x-8x^2}$

$$\Rightarrow y\alpha+6xy-8x^2y=\alpha x^2+6x-8$$

$$\Rightarrow (\alpha+8y)x^2+(6-6y)x-8-\alpha y=0$$

$x \in R$; $\quad D \geq 0$

$$(6(1-y))^2-4(\alpha+8y)(-8-\alpha y)\geq 0$$

$$4\left(9+9y^2-18y+8\alpha+\alpha^2y+64y+8\alpha y^2\right)\geq 0$$

$\Rightarrow y^2(9+8\alpha)+y(\alpha^2+46)+8\alpha+9\geq 0$

$\Rightarrow 9+8\alpha>0$ & $D\leq 0$

$\alpha>-\frac{9}{8}$ & $(\alpha^2+46)^2-4(8\alpha+9)^2<0$ _____(1)

$(\alpha^2+46+16\alpha+18)(\alpha^2+46-16\alpha-18)\leq 0$

$(\alpha^2+16\alpha+64)(\alpha^2-16\alpha+28)\leq 0$

$(\alpha+8)^2(\alpha-14)(\alpha-2)\leq 0$

$\alpha\in[2,14]\cup\{-8\}$ _____(2)

from (1) & (2)

$\alpha\in[2,14]$

for $\alpha=3$, $y=\frac{\text{Quad}}{\text{Quad}}$, many-one function.

13. The number of maps from the set {1, 2, 3} into the set {1, 2, 3, 4, 5} such that f(i) ≤ f(j) whenever i < j is _____

समुच्चय {1, 2, 3} से समुच्चय {1, 2, 3, 4, 5} में परिभाषित फलन f, इस प्रकार कि f (i) ≤ f(j) जब भी i < j की संख्या होगी _____

Ans. 35

Sol. Case-I When all the mappings are distinct

No. of functions $\Rightarrow {}^5C_3=10$

Case-II When two of the mappings are same

No. of functions $\Rightarrow {}^5C_2\times 2=20$

A factor of 2 has to be multiplied because after we have chosen two numbers, there can be two cases, aab or abb.

Case-III All mappings are same.

No. of functions = 5

Total functions = 35

14. Let the range of the function $f: \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ assumes exactly 3 distinct values. If the number of such function is N, then find the value of $\frac{N}{300}$.

माना फलन $f: \{1, 2, 3, 4, 5\} \to \{1, 2, 3, 4, 5\}$ के परिसर में ठीक तीन भिन्न मान है। यदि ऐसे फलनों की संख्या N है, तो $\frac{N}{300}$ का मान ज्ञात करो।

Ans. 5

Sol. ${}^5C_3$ (selecting 3 distinct range values) $\times$ $(3^5 - {}^3C_2\, 2^5 + {}^3C_1)$ (No. of onto functions) $= 1500 = N$

$$\frac{N}{300} = \frac{1500}{300} = 5$$

Paragraph for Question Nos. 15 to 17

Given a function $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\pi\left(\frac{(1-2[x])}{2}\right) + \sin\left(\frac{x-[x]}{2}\right)$ whose fundamental period is p, (where {.} and [.] represent fractional part and greatest integral part functions respectively) and $y = \sqrt{2p + \frac{p}{2}[x] - [x]^2}$ the domain of y is [q, r), one another function $f_2(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$ then on the basis of above information answer the following.

मूलभूत आवर्तकाल p का एक फलन $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\pi\left(\frac{(1-2[x])}{2}\right) + \sin\left(\frac{x-[x]}{2}\right)$ दिया गया है। (जहाँ {.} और [.] क्रमशः भिन्नात्मक भाग फलन तथा महत्तम पूर्णांक फलन को बताते है।) तथा $y = \sqrt{2p + \frac{p}{2}[x] - [x]^2}$ है एवं y का प्रान्त [q, r), हैं।

एक दूसरा फलन $f_2(x) = \begin{cases} 2+x, & x \geq 0 \\ 2-x, & x < 0 \end{cases}$

हो, तो दी गई सूचना के आधार पर निम्न के उत्तर दीजिए।

15. The period p of $f_1(x)$ is :

(A) An irrational number
(B) Prime number
(C) A composite number
(D) Neither prime nor a composite number

$f_1(x)$ का आवर्तकाल p है :

(A) एक अपरिमेय संख्या
(B) अभाज्य संख्या
(C) एक संयुक्त संख्या
(D) न तो अभाज्य न ही संयुक्त संख्या

Ans. C

16. Value of $r - q - 1$ is equal to :

$r - q - 1$ का मान है :

(A) 6 (B) 7 (C) 8 (D) 9

Ans. A

17. Range of $f_2(f_2(x))$ in terms of p, q, r is :

p, q, r के पदों में $f_2(f_2(x))$ का परिसर है :

(A) $[p, \infty)$ (B) $[q, \infty)$ (C) $[r, \infty)$ (D) $(-\infty, -r) \cup (p, \infty)$

Ans. A

Sol. $f_1(x) = e^{\tan\left\{\frac{x}{4}\right\}} + \cos\left(\frac{\pi}{2} - \pi[x]\right) + \sin\left(\frac{\{x\}}{2}\right)$

$e^{\tan\left\{\frac{x}{4}\right\}} \to T = 4$

$\cos\left(\frac{\pi}{2} - \pi[x]\right) = \sin(\pi[x]) = 0$

$\sin\left(\frac{\{x\}}{2}\right) \to T = 1$

$P = 4$

$\therefore y = \sqrt{8 + 2[x] - [x]^2}$

$8 + 2[x] - [x]^2 \geq 0$

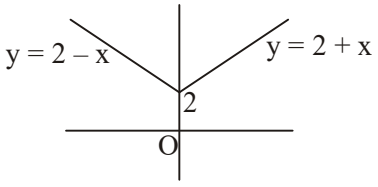
$([x] - 4)([x] + 2) \leq 0$

$[x] \in [-2, 4]$

$x \in [-2, 5)$

$q = -2, r = 5$

$\Rightarrow f_2(x) = \begin{cases} 2 + x; & x \geq 0 \\ 2 - x; & x < 0 \end{cases}$



$$f_2\left(f_2(x)\right)=\begin{cases}2+f_2(x); & f_2(x)\geq 0\\ 2-f_2(x); & f_2(x)<0\end{cases}$$

$$=\begin{cases}4-x; & x<0\\ 4+x; & x\geq 0\end{cases}$$

Range : $[4, \infty)$

Paragraph for Question Nos. 18 to 21

Let $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$. Let $f: (-\infty, a] \to [b, \infty)$, where 'a' is the largest real number for which $f(x)$ is bijective.

माना $f(x) = x^2 - 2x - 1 \ \forall \ x \in R$ है। माना $f: (-\infty, a] \to [b, \infty)$, जहाँ 'a' अधिकतम वास्तविक संख्या जिसके लिए $f(x)$ एकैकी आच्छादक है।

18. The value of $(a + b)$ is equal to :

$(a + b)$ का मान होगा :

(A) -2 (B) -1 (C) 0 (D) 1

Ans. B

Sol. $f(x) = (x-1)^2 - 2 \qquad a = 1, b = -2$

$a + b = -1$ Ans.

19. Let $f: R \to R$, $g(x) = f(x) + 3x - 1$, then the least value of function $y = g(|x|)$ is :

माना $f: R \to R$, $g(x) = f(x) + 3x - 1$, तो फलन $y = g(|x|)$ का न्यूनतम मान होगा :

(A) $-9/4$ (B) $-5/4$ (C) -2 (D) -1

Ans. C

Sol. $g(x) = x^2 + x - 2 = \left(x + \frac{1}{2}\right)^2 - \frac{9}{4} \qquad \Rightarrow \qquad g|x| = \left(|x| + \frac{1}{2}\right)^2 - \frac{9}{4}$

$g_{min} = g(0) = -2$ Ans.

20. Let $f: [a, \infty) \to [b, \infty)$, then $f^{-1}(x)$ is given by :

माना $f: [a, \infty) \to [b, \infty)$, तो $f^{-1}(x)$ होगा :

(A) $1 + \sqrt{x+2}$ (B) $1 - \sqrt{x+3}$ (C) $1 - \sqrt{x+2}$ (D) $1 + \sqrt{x+3}$

Ans. A

Sol. $f: [1, \infty) \to [-2, \infty)$, then $f^{-1}: [-2, \infty) \to [1, \infty)$;

$f(x) = y \quad \Rightarrow \quad x^2 - 2x - (1 + y) = 0$

$x = \dfrac{2 \pm \sqrt{4 + 4(1+y)}}{2}$; $\quad x = 1 \pm \sqrt{2+y}$; $\quad f^{-1}(y) = 1 + \sqrt{2+y}$;

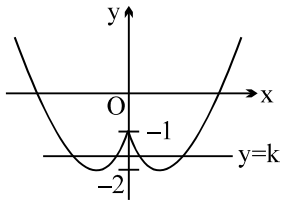
$f^{-1}(x) = 1 + \sqrt{2+x}$ Ans.

21. Let $f: R \to R$, then range of values of k for which equation $f(|x|) = k$ has 4 distinct real roots is :

माना $f: R \to R$, तो समीकरण $f(|x|) = k$ के 4 भिन्न हल होने के लिए k का परिसर होगा :

(A) $(-2, -1)$ (B) $(-2, 0)$ (C) $(-1, 0)$ (D) $(0, 1)$

Ans. A

Sol. 

; from graph $k \in (-2, -1)$ Ans.

MATCH THE COLUMN

22. Match the following columns :

Column – I	**Column – II**
(A) The period of the function $y = \sin(2\pi t + \pi/3) + 2\sin(3\pi t + \pi/4) + 3\sin 5\pi t$	(P) 1/2
(B) $y = \{\sin(\pi x)\}$ is a many one function for $x \in (0, a)$ where a may be	(Q) 8
(C) The fundamental period of the function $y = \dfrac{1}{2}\left(\dfrac{\lvert\sin(\pi/4)x\rvert}{\cos(\pi/4)x} + \dfrac{\sin(\pi/4)x}{\lvert\cos(\pi/4)x\rvert}\right)$	(R) 2
(D) If $f: [0, 2] \to [0, 2]$ is bijective function defined by $f(x) = ax^2 + bx + c$, where a, b, c are non-zero real numbers, then f(2) is equal to	(S) 0

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) निम्न फलन का आवर्तकाल $y = \sin(2\pi t + \pi/3) + 2\sin(3\pi t + \pi/4) + 3\sin 5\pi t$	(P) 1/2
(B) $x \in (0, a)$ के लिए $y = \{\sin(\pi x)\}$ एक बहुएकैकी फलन है, जहाँ a का मान हो सकता है	(Q) 8
(C) फलन $y = \dfrac{1}{2}\left(\dfrac{\lvert\sin(\pi/4)x\rvert}{\cos(\pi/4)x} + \dfrac{\sin(\pi/4)x}{\lvert\cos(\pi/4)x\rvert}\right)$ का मूलभूत आवर्तकाल	(R) 2
(D) यदि $f: [0, 2] \to [0, 2]$ एक एकैकी आच्छादक फलन हैं जो $f(x) = ax^2 + bx + c$ से परिभाषित है, जहाँ a, b, c अशून्य वास्तविक संख्याएँ हैं, तो $f(2) =$	(S) 0

Ans. (A) QR (B) QR (C) Q (D) S

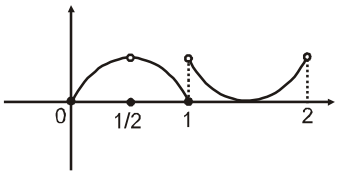
Sol. (A) $y = \sin\left(2\pi t + \frac{\pi}{3}\right) + 2\sin\left(3\pi t + \frac{\pi}{4}\right) + 3\sin(5\pi t)$

$$\text{LCM}\left(\frac{2\pi}{2\pi}, \frac{2\pi}{3\pi}, \frac{2\pi}{5\pi}\right) = \text{LCM} = \left(1, \frac{2}{3}, \frac{2}{5}\right) = 2$$

(B) $y = \{\sin \pi x\}$ period = 2

For $x \in \left(0, \frac{1}{2}\right)$ function is one-one

For $x \in (0, 2)$ and $x \in (0, 8)$
function is many-one



(C) $y = \frac{1}{2}\left(\frac{\left|\sin\frac{\pi}{4}x\right|}{\cos\frac{\pi}{4}x} + \frac{\sin\frac{\pi}{4}x}{\left|\cos\frac{\pi}{4}x\right|}\right) \Rightarrow$ F.T.P. of y is 8.

(D) Since f(x) is bijective,
$\therefore$ $f(0) = 0$ or 2 but $f(0) = 0 \Rightarrow c = 0$
(which is not true) $\therefore$ $f(0) = 2$ & $f(2) = 0$

23. Match the following columns :

Column – I	**Column – II**
(A) Domain of $f(x) = \sqrt{2^x - 3^x} + \log_{\frac{1}{2}} x$ is	(P) [0, 1]
(B) Solution set of equation $2\cos^2\frac{x}{2}\sin^2 x = x^2 + \frac{1}{x^2}$ is	(Q) $[0, \infty)$
(C) If $A = \{(x, y) ; y = \frac{1}{x}, x \in R_0\}$ and $B = \{(x, y); y = -x, x \in R\}$ then $A \cap B$ is	(R) $[1, \infty)$
(D) The functions $f(x) = \sqrt{x}\ \sqrt{x-1}$ and $g(x) = \sqrt{x^2 - x}$ are identical in	(S) ϕ

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) $f(x) = \sqrt{2^x - 3^x} + \log_{\frac{1}{2}} x$ का प्रान्त होगा	(P) [0, 1]

(B) समीकरण $2\cos^2\frac{x}{2}\sin^2 x = x^2 + \frac{1}{x^2}$ का हल होगा (Q) $[0, \infty)$

(C) यदि $A = \{(x, y);\ y = \frac{1}{x}, x \in R_0\}$ तथा $B = \{(x, y); y = -x, x \in R\}$ (R) $[1, \infty)$

तो $A \cap B$ होगा

(D) फलन $f(x) = \sqrt{x}\ \sqrt{x-1}$ तथा $g(x) = \sqrt{x^2 - x}$ किस अन्तराल में समरूप होंगे (S) ϕ

Ans. (A) S (B) S (C) S (D) R

Sol. (A) $2^x - 3^x \geq 0$

$2^x \geq 3^x$ $\qquad g(x) = \log_{\frac{1}{2}} x$

$x \in (-\infty, 0]$ $\qquad x > 0$

Domain of $f(x)$ is ϕ

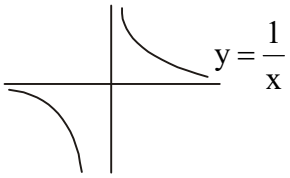
(B) $f(x) = (1 + \cos x)\sin^2 x = (1 + \cos x)(1 - \cos^2 x)$

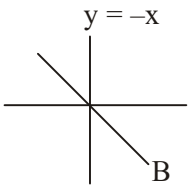
$= 1 + \cos x - \cos^2 x - \cos^3 x$

Put $\cos x = t; \Rightarrow 1 + t - t^2 - t^3$

for $t \in [0,1]$; $1 + t - t^2 - t^3 \in [0,1]$

$\therefore$ LHS $\in [0,1]$; RHS $[2, \infty]$

(C) 



$A \cap B = \phi$

(D) Domain of $f(x)$ & $g(x)$ is $[1, \infty)$

24. Match the following columns :

Column – I	**Column – II**
(A) The function $f(x) = \frac{3x-2}{x+4}$ has an inverse that can be written in the form $f^{-1}(x) = \frac{x+b}{cx+d}$. The value of $(b + c + d)$ is	(P) even

(B) Let $f(x)=\frac{x}{1+x}$ and let $g(x)=\frac{rx}{1-x}$. Let S be the set of all real numbers r such that $f(g(x))=g(f(x))$ for infinitely many real number x. The number of elements in set S is — (Q) odd

(C) If $f(x)=2x+1$ then the value of x satisfying the equation $f(x)+f(f(x))+f(f(f(x)))+f(f(f(f(x))))=116$, is — (R) prime

(S) neither prime nor composite

निम्न कॉलम को सुमेलित कीजिए :

Column – I	**Column – II**
(A) फलन $f(x)=\frac{3x-2}{x+4}$ का व्युत्क्रम $f^{-1}(x)=\frac{x+b}{cx+d}$ है, तो (b + c + d) का मान होगा	(P) सम
(B) माना $f(x)=\frac{x}{1+x}$ तथा let $g(x)=\frac{rx}{1-x}$. माना S सभी वास्तविक संख्याओं r का समुच्चय इस प्रकार है कि $f(g(x))=g(f(x))$ बहुत सारे अनन्त वास्तविक x के लिए है, तो समुच्चय S में अवयवों की संख्या होगीं	(Q) विषम
(C) यदि $f(x)=2x+1$ है, तो समीकरण $f(x)+f(f(x))+f(f(f(x)))+f(f(f(f(x))))=116$ को संतुष्ट करने वाले x का मान होगा।	(R) अभाज्य
	(S) ना अभाज्य और ना ही संयुक्त

Ans. (A) QS (B) P R (C) QR

Sol. (A) $y=\frac{3x-2}{x+4}$

$xy+4y=3x-2$

$x(y-3)=-4y-2$

$$x=\frac{-4y-2}{y-3}=\frac{y+\frac{1}{2}}{\frac{3}{4}-\frac{1}{4}y}$$

$$f^{-1}(x)=\frac{x+\frac{1}{2}}{\frac{3}{4}-\frac{1}{4}x}$$

$$b=\frac{1}{2},c=-\frac{1}{4},d=\frac{3}{4}$$

(B) $f(g(x)) = g(f(x))$

$$\frac{\frac{rx}{1-x}}{1+\frac{rx}{1-x}} = \frac{r\frac{x}{1+x}}{1-\frac{x}{1+x}}$$

$$\Rightarrow \frac{rx}{1-x+rx} = rx$$

$r = 0$ or 1

(C) $f(x) = 2x + 1$

$f(f(x)) = 2(2x + 1) + 1 = 4x + 3$

$f(f(f(x))) = 2(4x + 3) + 1 = 8x + 7$

$f(f(f(f(x)))) = 2(8x + 7) + 1 \Rightarrow 16x + 15$

$\Rightarrow (2x + 1) + (4x + 3) + (8x + 7) + (16x + 15) = 116$

$\Rightarrow 30x + 26 = 116$

$30x = 90$

$x = 3$